Journal of Sound and Vibration 461 (2019) 114906

Contents lists available at ScienceDirect

Journal of Sound and Vibration

journal homepage: www.elsevier.com/locate/jsvi

Development and validation of a simple two degree of freedom model for predicting maximum fundamental sloshing mode wave height in a cylindrical tank

Vikas Sharma [b], C.O. Arun [a], I.R. Praveen Krishna [a,*]

[a] Department of Aerospace Engineering, Indian Institute of Space Science & Technology, Valiamala, Thiruvananthapuram, 695547, India
[b] Liquid Propulsion Systems Center, Indian Space Research Organization (ISRO), Thiruvananthapuram, 695547, India

ARTICLE INFO

Article history:
Received 26 March 2019
Revised 7 August 2019
Accepted 14 August 2019
Available online 20 August 2019
Handling Editor: L. Huang

Keywords:
Maximum sloshing wave height
2 DOF model
Beats
CFD
Mechanical model

ABSTRACT

The present paper proposes a new and simple mechanical model for predicting the maximum sloshing wave height (MSWH) in a partially filled, laterally excited cylindrical tank. The proposed model is a linear, two degree of freedom spring-mass-damper system which consists of two masses representing the sloshing and non-sloshing mass of the partially filled cylindrical tank. The model is validated through a series of experiments conducted on a cylindrical tank filled with water which is mounted in a slosh test rig. The proposed model successfully captures the beating phenomenon observed in sloshing wave height, when the tank is laterally excited near the resonant frequency of the liquid. The present study leads to a simpler, accurate and faster mechanical model which can predict the MSWH, frequency of sloshing liquid and the lateral sloshing force. The model can be readily used by the design engineers, instead of time consuming numerical models like computational fluid dynamics and smoothed particle hydrodynamics, in the preliminary design phase of any structure involving lateral sloshing and provide quick results which will substantially lead to shorter design cycle time.

1. Introduction

Sloshing refers to the oscillatory motion of the free surface of a liquid in a partially filled container, when the container is excited. The physics behind sloshing of liquid is complicated due to the non-linearities associated with the free surface of the liquid. The sloshing phenomenon can be observed in widespread applications like automobiles, liquid storage tanks, aircrafts and space launch vehicles. Sloshing of fuel in road tankers while turning manoeuvres can lead to fatal accidents. Space fairing rockets carry 90% of their initial weight as propellants and these can cause significant amount of disturbing side forces on the launch vehicle due to sloshing. Also, the vibratory motion of propellant caused due to sloshing can couple with the structural vibration of the launch vehicle and result in catastrophic failures. The ground excitations induced by an earthquake can also trigger sloshing in the liquid storage tanks which can further lead to significant damages to the tank.

Hatayama [1] studied the damage caused to the oil storage tanks in Japan by the Tohuku earthquake in 2011. Liquid sloshing, in all the preceding examples, is found to be an undesirable phenomenon which needs to be quantified and managed. On the other hand, there are applications which purposefully induce sloshing to achieve desirable results. One such classic example is sloshing vibration absorbers or tuned sloshing dampers (TSD), which are very effective in controlling structural vibrations.

* Corresponding author.
E-mail address: praveenkrishna@iist.ac.in (I.R.P. Krishna).

https://doi.org/10.1016/j.jsv.2019.114906

Yamamoto and Kawahara [2] studied the effect of employing tuned liquid dampers to control the structural vibrations of the Yokohama marine tower. The preceding examples emphasize the importance of sloshing analysis.

The existing mechanical models of sloshing such as pendulum and spring-mass-damper models [3], focus only on predicting the magnitude of forces and moments caused by the sloshing liquid on the side walls of the container. The beating nature of sloshing force is not considered in these models and is assumed to be purely sinusoidal in nature. Because of the above assumption, the maximum sloshing wave height (MSWH) also becomes purely sinusoidal. However, the experimentally observed MSWH is not purely sinusoidal and is observed to show beats. The sloshing wave of liquid can impact the roof of the liquid storage tanks during an earthquake and can cause significant damages [4]. During the burnout phase of liquid rocket, the sloshing of propellant inside the tank, can expose the tank outlet to gaseous propellant and can cause cavitation in the turbopump of the liquid engine. In addition, sufficient ullage volume is kept in the propellant tanks, to accommodate the sloshing liquid. Hence, the prediction of MSWH is equally important in sloshing study.

Researchers have analysed the problem of liquid sloshing in different perspectives. Some of the works in the field of slosh studies were done in early nineteen hundreds [5–9]. The basic theory behind the moving free surface of a liquid in a partially filled container was investigated by Lamb [10] and Stoker [11]. The boundary conditions and the equation of motion of incompressible, inviscid water waves with a free surface were derived in both Eulerian and Lagrangian frame of reference. These studies formed the basis for all further works in the area of slosh studies. The natural frequency of an inviscid and irrotational liquid in a circular cylindrical tank was derived by Miles [12] using the linear hydrodynamic theory. The expression for the damping ratio of the liquid and pressure distribution on the lateral wall of the tank were also estimated. However, the frictional force was entirely neglected in this study. Bauer and Eidel [13] obtained the natural damped frequencies and the response to lateral excitation of a viscous and incompressible liquid in a cylindrical container. This study concluded that the damping of liquid during sloshing was contributed by the free surface of the liquid, side wall, and bottom wall of the container. For the case of low liquid depth, i.e. when fluid depth is less than the radius of the container, major contribution to damping was observed from the bottom wall of the container. A detailed review of works done in the early 20th century on theoretical and experimental research in the area of sloshing can be found in Cooper [14].

The research in the area of sloshing accelerated with the surge in development of rockets during the second World War. The control and stability analysis of rocket, during its ascent thorough the atmospheric regime, necessitated the inclusion of sloshing dynamics in the global mathematical model of the rocket. Henceforth, many researchers [3,15–18] worked on the equivalent mechanical model which could represent the sloshing in containers of different shapes under different forms of excitation. Abramson et al. [3] have derived the equivalent mechanical model in the form of a simple spring-mass-damper, which was based on the principle of equality of forces and moments. The model was validated with already existing experimental data for sloshing in cylindrical tank during translation and pitching. However, this model does not address the sloshing wave height attained by the liquid during excitation. The sloshing force predicted is compared only for magnitude. The waveform of MSWH as well as the sloshing force was assumed to be purely harmonic. Warner and Caldwell [19] presented a detailed comparison of analytical and experimental results of frequencies and sloshing wave heights of mercury filled in a cylindrical tank during translation and pitching excitation. Only low amplitude planar oscillations of the liquid were considered in this study. They developed a pendulum based model for predicting the MSWH. The wave height of the sloshing liquid obtained from the pendulum model was compared with the experimentally obtained wave height. Damping was totally neglected in this model and therefore, only a qualitative comparison of sloshing wave height was provided. In 1966, Abramson [20] compiled a monograph, on sloshing of liquid in moving containers. This monograph consolidated all the earlier work published by various researchers on different aspects of sloshing.

Sloshing, as mentioned earlier, has been purposefully induced in systems to achieve vibration control of structures and is reported in many research papers [21–24]. Yalla and Kareem [21] studied the response of structure using a tuned liquid column damper (TLCD) systems. A U-tube containing oscillating liquid, which acts as TLCD to control the vibration of the structure, was mounted on top of the structure. The entire system was modeled as a two degrees of freedom spring-mass-damper model whose equations of motion were derived using Newton's second law of motion and were solved numerically. The response of vibrating structure was found to be in the form of beats which disappeared when a sloshing liquid of high damping ratio was used. The beats observed were attributed to the mass coupling present in the system. However, this paper did not discuss the coupling parameter and a trial and error method was used to get the response. Being a passive vibration control study the focus of the work was fully on the vibration of the structure. The nature and the magnitude of MSWH of TLCD were not examined. Also no systematic methodology was mentioned in the paper to find the optimum coupling parameter of the given system.

The energy dissipated in a sloshing vibration absorber was studied by Cavalagli et al. [22]. A partially filled rectangular tank was modeled as a single degree of freedom (SDOF), linear spring-mass-damper system. This model was different from the mechanical model proposed by Abramson et al. [3] in the sense that, this model was based on the equivalence of energy dissipation. Here, the liquid was assumed to be incompressible and the tank velocities were subsonic. In addition to the analytical model, a rectangular tank partially filled with water was used for an experimental study. The tank was laterally excited at different frequencies. Shear force between the tank and moving table was used for calculating the energy dissipated in the liquid. This experimental study revealed that the maximum energy dissipation in the liquid occurred at excitation frequencies much higher than the sloshing frequency which was not captured in the mechanical model. On the other hand, Computational Fluid Dynamics (CFD) simulations using OpenFOAM software package could clearly capture the experimental results which demonstrated the superiority of CFD model over a simple mechanical model in determining the energy dissipated in a sloshing vibration absorber. However, the CFD simulation is very tedious, expensive and requires expertise to carry out analysis. Moreover, the model proposed by Cavalagli et al. [22] did not discuss the wave height attained by the sloshing liquid and it also excluded non-sloshing mass of liquid from the model.

Anderson et al. [23] presented a sloshing vibration absorber as an alternative to a tuned vibration absorber for controlling the vibration of a structure. Unlike the U-tube TLCD proposed by Yalla and Kareem [21] for vibration damping, the TSD proposed here was in the form of a rectangular tank partially filled with water. This was mounted on the top of the structure whose vibratory motion is to be controlled. The work suggests a 2 DOF system; one freedom belonging to the structure to be controlled, second the sloshing liquid in its fundamental standing wave mode. Hence, the liquid a sdof representation. However they used CFX (a commercial CFD package) to model the rectangular tank (as a two dimensional numerical model) to find the response of free surface. In this paper, the author had successfully predicted the beat phenomenon observed in the structural motion and concluded that there exists an optimal value of damping factor of the sloshing liquid for which the damping control was found to be most effective. This study concentrated on the motion of structure alone and did not discuss the wave motion of liquid inside TSD.

Goudarzi and Yazdi [25] had proposed a CFD based method, in addition to an analytical method, to obtain the MSWH in a partially filled rectangular tank which was laterally excited. The analytical model was based on the theory proposed for inviscid and irrotational liquid in a rectangular tank by Wu et al. [26], which considered the MSWH to be composed of two terms, called as "wave train". One term corresponded to the excitation frequency of the container and another term to the natural frequency of the liquid. A non-linear model, based on the Volume of Fluid method, was implemented in CFD to study the non-linearities associated with the free surface when the tank containing liquid was excited near the resonant frequency. The work was validated experimentally in a rectangular tank, filled with water up to different heights and excited sinusoidally. The MSWH obtained from the CFD model compared well with experimental results for all excitations frequencies. On the other hand, the analytical model failed to capture the MSWH near the resonant frequency of liquid.

In all the preceding papers related to use of sloshing absorbers, the emphasis has been on controlling the vibratory motion of the structure due to external disturbances. The response of the sloshing liquid does not find any mention in these studies. The spring-mass-damper model suggested by Abramson et al. [3] concentrated on the prediction of forces and moments exerted by the sloshing liquid. The discussion about the beat phenomenon observed in the sloshing wave height, when the excitation frequency is in the vicinity of the natural frequency of sloshing liquid, was found to be lacking. Computational methods (CFD, Smoothed Particle Hydrodynamics (SPH) etc.) can be used to predict the free surface wave height of sloshing liquid; however they are complex, tedious and expensive to implement. They also needs expertise to implement correctly. Furthermore, the analytical solution proposed by Wu et al. [26] to estimate the MSWH, is applicable only for a rectangular tank. A systematic procedure for finding out the parameters like stiffness and damping of sloshing or nonsloshing mass of liquid; to be used in the simple mechanical model, is missing in the current literature. This is very essential for the design engineers to generate simple models of their systems. In this scenario, the present paper proposes a newer and simple spring-mass-damper model to predict the MSWH of the free surface in a partially filled cylindrical tank undergoing lateral sloshing. A detailed explanation of the parameters to be used in the mechanical model and their evaluation methodology is also discussed in this paper. The proposed model is a two degree of freedom, spring-mass-damper system solved numerically by RKF45 (Runge-Kutta-Fehlberg) scheme. The proposed mathematical model is validated experimentally using a slosh test rig.

2. Theory of lateral sloshing in cylindrical tank

Lateral sloshing refers to the response of the liquid surface in an upright container, subjected to a lateral excitation. Lateral sloshing assumes importance due to the fact that it produces asymmetric mode shapes. The theory of liquid sloshing in a cylindrical tank without baffles has been discussed in many research papers [12,20,27–29]. A brief discussion about the governing equations and boundary conditions pertaining to the present problem of lateral sloshing in a cylindrical tank is provided here. In general, a sloshing problem can be modeled in two different ways. One method consists of using the linearized hydrodynamic theory [16] which is based on the velocity potential function. This theory is used for the derivation of the natural frequency of the sloshing liquid. The other method [20] represents the sloshing problem using equivalent mechanical models. The following sections briefly review these two approaches.

2.1. Linearized hydrodynamic theory of liquid sloshing

When a partially filled container undergoes lateral excitation, the free surface of the liquid column inside the container may assume arbitrary shapes such as planar, non-planar, beating or chaotic [27]. The theoretical analysis of such a problem can be very difficult due to the nature of the free surface acquired by the sloshing liquid. Therefore, the following assumptions [20] are invoked to simplify this problem.

1. The tank containing the liquid is assumed to be rigid to avoid the complexities associated with the flexibility of container modes.
2. The free surface wave heights are assumed to be small in order to linearize the problem.
3. The liquid is assumed to be incompressible, irrotational, homogeneous and inviscid.
4. The total volume of liquid inside the tank is constant.

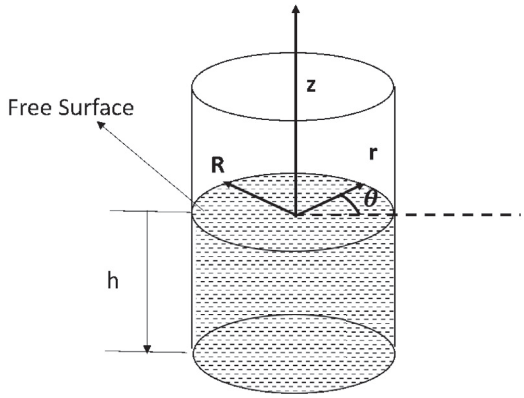

Fig. 1. Cylindrical coordinate system (r, θ, z) assumed for the cylindrical tank containing liquid up to depth of h.

A cylindrical tank of radius, R, partially filled with liquid up to a height, h, is considered. A cylindrical coordinate system (r,θ,z), with the origin at the center of the liquid free surface, is employed (see Fig. 1).

Let φ represent the single valued velocity potential function, then the governing equation for the liquid can be written as,

$$\nabla^2 \varphi = 0 \tag{1}$$

The normal velocity of liquid particles, at the tank bottom as well as at the tank periphery, must be zero. This leads to the following boundary conditions,

$$\left.\frac{\partial \varphi}{\partial z}\right|_{z=-h} = 0 \tag{2}$$

and,

$$\left.\frac{\partial \varphi}{\partial r}\right|_{r=R} = 0 \tag{3}$$

The kinematic free surface condition, that particles of the surface must remain on the surface, with the condition that pressure on the surface of the liquid is constant, gives

$$\frac{\partial^2 \varphi}{\partial t^2} + g\frac{\partial \varphi}{\partial z} = 0, \tag{4}$$

where g is the acceleration due to gravity and t is time. The velocity potential function [27] satisfying Eq. (1), along with the boundary conditions defined in Eqs. (2) and (3), can be obtained as,

$$\varphi = \sum_m \sum_n \left[\alpha_{mn}(t)\cos(m\theta) + \beta_{mn}(t)\sin(m\theta)\right] J_m\left(\frac{\xi_{mn} r}{R}\right) \frac{\cosh[\frac{\xi_{mn}}{R}(z+h)]}{\cosh(\xi_{mn}h/R)} \tag{5}$$

where α_{mn}, β_{mn} are time dependent variables, $J_m(.)$ is the Bessel function of first kind and ξ_{mn} are the roots of Bessel function. Here, m represents the mth mode of vibration in tangential direction and n represents the nth mode of vibration in radial direction.

Expressing α_{mn} and β_{mn} as harmonics functions, and substituting Eq. (5) in the free surface condition given in Eq. (4), the natural frequency [27], ω_{mn}, of the liquid column is obtained as,

$$\omega_{mn}^2 = \frac{g\xi_{mn}}{R}\tanh(\xi_{mn}h/R) \tag{6}$$

The value of ξ_{mn} [30] for the fundamental mode ($m = 1, n = 1$) is found to be 1.84. Hence, the natural frequency of the liquid column, considering only the fundamental mode, is given by

$$\omega_{11}^2 = \frac{1.84g}{R}\tanh(1.84h/R) \tag{7}$$

2.2. Existing mechanical model for sloshing

The lateral sloshing problem can also be represented using equivalent mechanical models like the spring-mass-damper system [3] and the pendulum based models [20]. The parameters of the spring-mass-damper model, proposed by Abramson et al. [3], is based on the following assumptions; (i) Equivalent masses and moment of inertia between the model

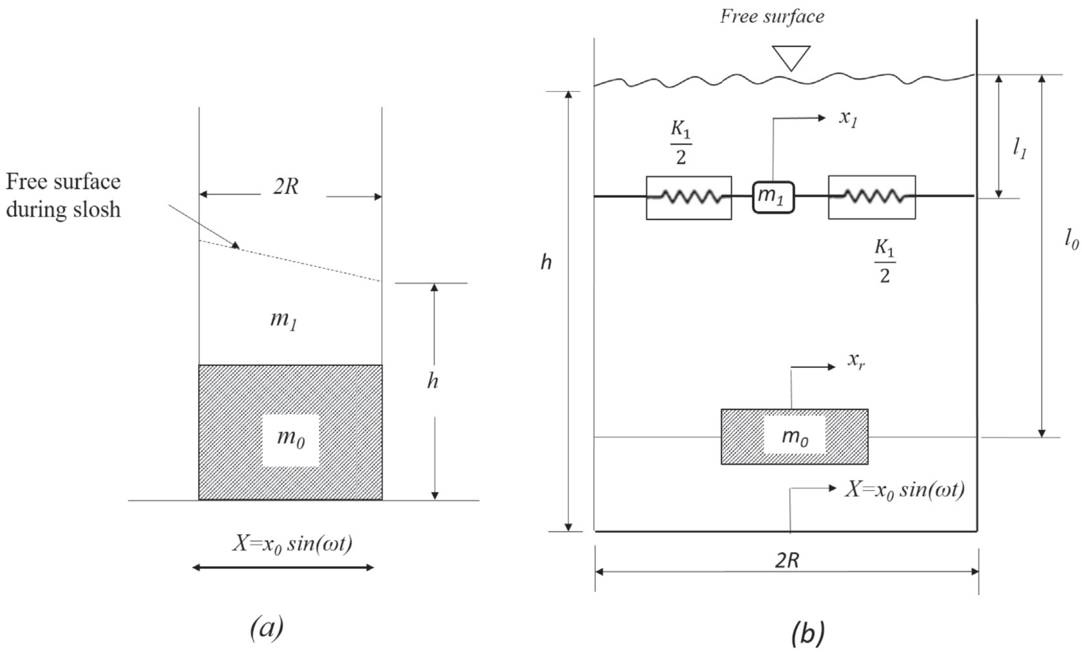

Fig. 2. (a) Cylindrical tank during lateral sloshing (b) Spring mass model representing the sloshing of liquid in a cylindrical container during lateral slosh [20].

and the sloshing liquid must be preserved. (ii) The model must produce the same damping forces as the sloshing liquid. (iii) For small oscillations, the center of gravity of the model remains the same. (iv) Forces and moments produced by the sloshing liquid, which can be obtained using the linear hydrodynamic theory, must be equal to those obtained from the model.

Fig. 2(a) shows sloshing liquid in a cylindrical container during lateral excitation. This sloshing liquid is represented by a series of spring-mass-damper elements including a rigid mass.

Fig. 2(b) shows the simplest spring mass model proposed by Abramson et al. [20,27] which assumes zero damping and considers only the fundamental modal mass, m_1 [20,27]. The total liquid is represented using two masses, m_0 and m_1. The lower mass, m_0, acts as a rigid mass which does not participate in sloshing and moves with an acceleration same as that of the container. The upper mass, m_1, is referred to as sloshing mass which participates in sloshing. K_1 is the stiffness of the sloshing mass, m_1. l_0 and l_1 are the distances of rigid and modal masses from the free surface respectively. Let X be a harmonic excitation input to the model and x_r and x_1 be the responses of rigid and sloshing mass respectively.

The total liquid mass, M, in the tank is,

$$M = m_0 + m_1 \tag{8}$$

The equation of motion for the sloshing mass is,

$$m_1\ddot{x_1} + K_1 x_1 = m_1 x_0 \omega^2 \sin(\omega t) \tag{9}$$

The parameters of this model can be described as follows.

The natural frequency of the sloshing mass is

$$\omega_{11}^2 = \frac{K_1}{m_1} \tag{10}$$

From the above model, lateral force acting on the walls of the tank by sloshing liquid due to a pure translation excitation, $X = x_0 \sin(\omega t)$, is given by Ref. [27],

$$F_x = M x_0 \omega^2 \sin(\omega t)\left\{1 + \frac{m_1}{M}\left(\frac{\omega^2}{\omega_{11}^2 - \omega^2}\right)\right\} \tag{11}$$

Furthermore, the linearized hydrodynamic theory of sloshing [27] derives the lateral force exerted by the sloshing mass on the tank wall as,

$$F_x = M x_0 \omega^2 \sin(\omega t)\left\{1 + \frac{2R\omega^2 \tanh(1.84h/R)}{4.4h(\omega_{11}^2 - \omega^2)}\right\} \tag{12}$$

Comparing Eqs. (11) and (12), the ratio of first modal mass to total liquid mass is obtained as,

$$\frac{m_1}{M} = \frac{R\tanh(1.84h/R)}{2.2h} \tag{13}$$

Also, the ratio of stiffness of fundamental modal mass to total mass [27] is obtained by substituting Eqs. (7) and (13) in Eq. (10) as,

$$\frac{K_1}{M} = \frac{g\{\tanh(1.84h/R)\}^2}{1.19h} \tag{14}$$

Eqs. (13) and (14) represents the parameters of spring mass model put forward by Abramson et al. [3] for lateral sloshing in cylindrical tank. The response of the sloshing mass, x_1, can be obtained by Eq. (15). The response is purely sinusoidal and being a SDOF system, the model fails to predict the observed beat phenomena, when the excitation frequency, ω, is in the vicinity of natural frequency, ω_{11}. This is the major drawback of this model.

$$x_1 = \frac{\omega^2 x_0 \sin(\omega t)}{\omega_{11}^2 - \omega^2} \tag{15}$$

3. Proposed two DOF spring-mass-damper model

3.1. Model description

A simple, two degree of freedom, spring-mass-damper system (Fig. 3) is used for representing the tank sloshing problem shown in Fig. 2a. Unlike the mechanical model proposed by Abramson et al. [3], m_0, in the proposed mechanical model includes the mass of tank and the mass of liquid which according to Abramson et al. does not participate in sloshing. Further, in the present study it is assumed that m_0 will influence the sloshing wave heights leading to a two degree of freedom system. The natural frequency of non-sloshing system is far higher than the sloshing part. This makes the relative movement of sloshing mass depend on the sweep frequency as well thus creating a beat pattern. m_1 represents the sloshing mass which is the top portion of the liquid column participating in sloshing. K_1 and C_1 are stiffness and damping coefficients of sloshing mass of liquid as explained by Abramson et al.. Additionally, in the proposed model, K_0 and C_0, are incorporated to represent the stiffness and damping of m_0 respectively. x_r is the response of m_0 and x_1 is the response of the sloshing mass of liquid which represents the MSWH.

The equations of motion for the proposed model, shown in Fig. 3, are derived using Newton's second law of motion as,

$$m_0\ddot{x}_r + K_0(x_r - X) + C_0(\dot{x}_r - \dot{X}) - K_1(x_1 - x_r) - C_1(\dot{x}_1 - \dot{x}_r) = 0 \tag{16}$$

$$m_1\ddot{x}_1 + K_1(x_1 - x_r) + C_1(\dot{x}_1 - \dot{x}_r) = 0 \tag{17}$$

For the steady state, Eqs. (16) and (17), together, can be written in matrix form as

$$\begin{bmatrix} m_0 & 0 \\ 0 & m_1 \end{bmatrix}\begin{bmatrix} \ddot{x}_r \\ \ddot{x}_1 \end{bmatrix} + \begin{bmatrix} C_0 + C_1 & -C_1 \\ -C_1 & C_1 \end{bmatrix}\begin{bmatrix} \dot{x}_r \\ \dot{x}_1 \end{bmatrix} + \begin{bmatrix} K_0 + K_1 & -K_1 \\ -K_1 & K_1 \end{bmatrix}\begin{bmatrix} x_r \\ x_1 \end{bmatrix} = \begin{bmatrix} K_0 X + C_0\dot{X} \\ 0 \end{bmatrix} \tag{18}$$

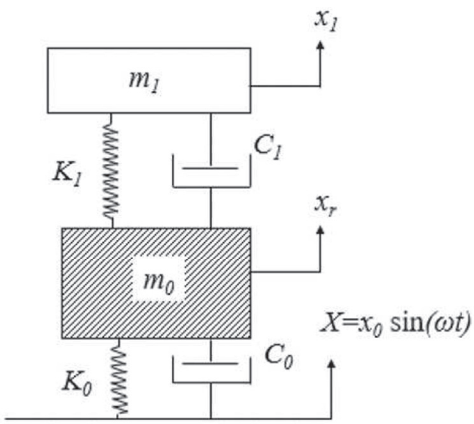

Fig. 3. Proposed two DOF spring-mass-damper model for lateral sloshing.

Table 1
Parameters of the sloshing mass, $\boldsymbol{m_1}$.

Height(m)	Sloshing mass (kg)	Stiffness, (N/m)	Damping, (Ns/m)
h	m_1	K_1	C_1
0.2	10.86	936.50	0.52
0.3	11.33	1019.47	0.55
0.4	11.41	1033.30	0.56
0.5	11.42	1035.51	0.56

This represents a very simple coupled system of equations which can be solved by numerical integration. The lateral force exerted by the sloshing liquid on the tank wall can be obtained from,

$$F_x = K_1 x_1 + C_1 \dot{x}_1 \tag{19}$$

In this equation, the stiffness parameters (which causes coupling) has physical meaning and one can estimate the same. One can also observe that even though the slosh force is due to the relative motion of the free surface in the container, in the model it is calculated from the absolute motion. This formulation helps the designer, to avoid using the trial and error method for finding the optimum parameters of the model and thus, in reducing the design cycle time. The following section explains the procedure for estimating the parameters of the proposed model.

3.2. Parameter estimation of sloshing mass, $\boldsymbol{m_1}$

The sloshing mass, m_1, and its stiffness, K_1, are computed using Eqs. (13) and (14) respectively. Damping, C_1, of the sloshing mass is computed as,

$$C_1 = \zeta_1 \times 2\sqrt{m_1 K_1} \tag{20}$$

where ζ_1 is the damping ratio of sloshing mass, m_1. The damping ratio, ζ_1, for sloshing liquid in circular cylindrical tank with flat bottom had been estimated empirically by conducting experiments and is provided in literature [20,31], as

$$\zeta_1 = 4.98\nu^{1/2}R^{-3/4}g^{-1/4}\left[1 + \frac{0.318}{\sinh\left(1.84\frac{h}{R}\right)}\frac{1-\frac{h}{R}}{\cosh\left(1.84\frac{h}{R}\right)})\right] \tag{21}$$

where ν is the kinematic viscosity of the fluid. For depths, $h > R$, Eq. (21) can be approximated as

$$\zeta_1 = 4.98\nu^{1/2}R^{-3/4}g^{-1/4} \tag{22}$$

This is due to the fact that the terms in the bracket reduces to unity as $\frac{h}{R}$ is greater than unity.

Table 1 summarizes the parameters of the sloshing mass for $R = 200$ mm. It can be observed from Table 1 that as the fill height is increased, the sloshing mass, as well as its stiffness is converging to a single value and damping, C_1, of the sloshing mass, is almost same for all fill heights. It can be observed that the change in slosh frequency is not heavily depends on the fill depth.

3.3. Parameter estimation of $\boldsymbol{m_0}$

The mass m_0, is composed of the non-sloshing mass of liquid and the mass of tank. Hence, m_0 is computed as,

$$m_0 = M_t + M - m_1. \tag{23}$$

Here, M is the total mass of liquid inside the tank, M_t is the mass of tank and m_1 is the sloshing mass. The sloshing mass and slosh damping values can be calculated as described in the above section.

In order to evaluate the support stiffness and damping (which is due to the tank and the nonsloshing mass), there are no analytical expressions available. These stiffness and damping parameters of m_0 are obtained from hammer impact test conducted on a stationary cylindrical tank in empty and filled condition.

This test setup consists of a steel cylindrical tank mounted on the test bench of a slosh test rig. The test is conducted first on the empty tank and then repeated sequentially for fill heights, 0.2 m–0.5 m, in steps of 0.1 m. The response of the tank during the impact is measured with an accelerometer(Kistler 8632C5) pasted on the front face of the tank. The data from the accelerometer is acquired with a VibPilot signal analyzer at a sampling rate of 2048 Hz in a Windows®based computer system.

Fig. 4 represents the Fast Fourier Transform (FFT) spectrum of the accelerometer response, showing the structural frequencies of the tank in empty as well as the filled condition. The most predominant frequency of 304.2 Hz is chosen as the natural frequency of the tank. The other predominant frequencies can be attributed to the support structure modes of the experimental setup. In order to verify the same, a finite element model of the empty tank alone is constructed in ANSYS®and modal analysis is carried out and which also shows the predominant natural frequency as 303.7 Hz. Further it is observed from Fig. 4 that the

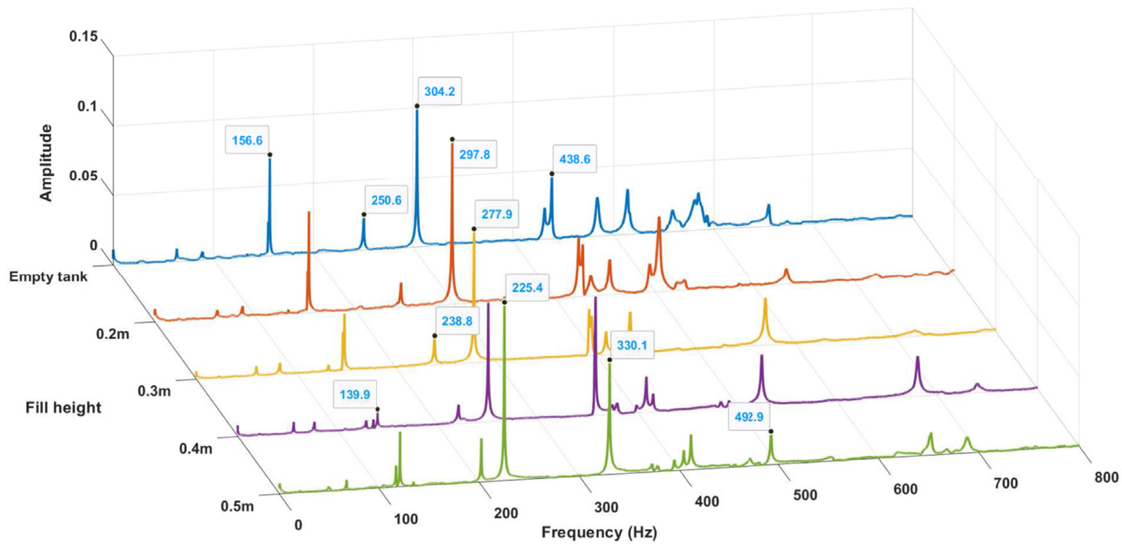

Fig. 4. Comparison of FFT responses of hammer impact test conducted on the tank in empty and filled condition.

Table 2
Parameters of m_0.

Height (m)	Total mass kg	Mass of tank + non-sloshing liquid, kg	Structural Frequency (Hz)	Stiffness N/m	Damping Ns/m
h	$M_t + M$	m_0	f_0	K_0	C_0
0	21.10	21.10	304.2	7.71×10^7	86.89
0.2	50.13	39.27	297.8	1.62×10^8	228.48
0.3	62.70	51.37	277.9	1.79×10^8	273.56
0.4	75.27	63.86	250.7	1.77×10^8	362.38
0.5	87.83	76.41	225.3	1.68×10^8	382.73

natural frequency is reduced to 297.8 Hz due to the addition of water up to 0.2 m fill height. The natural frequency of tank (f_0) is found to be reduced from 297.8 Hz to 225.3 Hz as the tank gets filled from 0.2 m to 0.5 m fill level due to the addition of mass in the form of water. In the impact experiments the frequency of sloshing mass is not considered as the free surface will have the influence of many higher modes. It is also to be noted that the slosh frequencies of interest are much lower than the structural frequencies. So their interaction with the tank modes are minimal. However, the damping of the system changes as the response of the tank (in the predominant fundamental mode) depends on the fluid damping as well. The change in support stiffness due to addition of water is very minimal as one can see from Table 2, as expected. The stiffness, K_0, of m_0 can be computed as,

$$K_0 = m_0(2\pi f_0)^2, \tag{24}$$

The damping ratio, ζ_0, of m_0 is calculated using the log decrement method. Therefore, the damping, C_0, of m_0 is computed as,

$$C_0 = \zeta_0 \times 2\sqrt{m_0 K_0} \tag{25}$$

Model parameters of m_0 are listed in Table 2.

4. Experimental setup

Experimental studies are conducted, in a slosh test rig, to validate the proposed model by comparing the MSWH of liquid in a cylindrical tank during lateral excitation. The test-setup schematic, shown in Fig. 5, mainly consists of a slosh test rig for providing excitation to the cylindrical tank, a data acquisition system for acquiring the MSWH of liquid and a vibration signal analyzer for acquiring the response of tank during excitation.

The slosh test rig consists of an electromechanical structure, test-bench, linear and rotary axis drive assembly, a receptacle for receiving the test article and a personal computer (PC). The test-bench is supported on both sides by spindle housings. The test-rig has its own built-in programmable logic controller (PLC) based control system. A specially designed pivot with anti-friction bearings is used for providing the desired movement to the test bench. Linear axis drive and rotary axis drive provide the linear and rotary movement to the test bench. For the present study, only pure lateral displacement is considered. A linear carriage houses the linear motor and gear head assembly which drives a servo actuator to move the test bench in an orthogonal direction

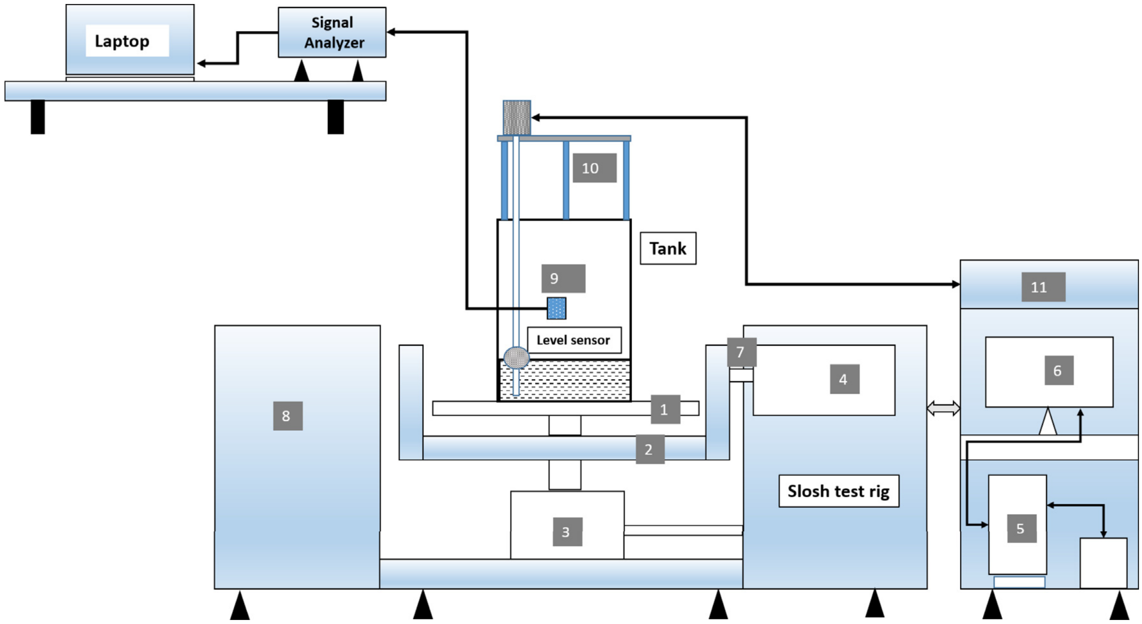

Fig. 5. Schematic diagram of the test setup for estimating the parameters of m_0 1. Test bench 2. Receptacle 3. Linear axis drive 4. Rotary axis drive 5. Personal Computer 6. Monitor 7. Spindle housing 8. Electromechanical structure 9. Accelerometer 10. Cross fixture for mounting the level sensor 11. Data Acquisition system for acquiring MSWH data.

with respect to the machine frame. A maximum linear displacement of about ±10 mm at a maximum frequency of 4 Hz can be provided to the test bench. Trapezoidal, sinusoidal and triangular wave forms can be provided as input to the test bench. The control system consists of a PC, high speed data acquisition system and a Windows®based software for issuing commands to the test-rig. The test rig has a mechanical stopper arrangement to prevent displacement going beyond the specification of the bench.

A stainless steel cylindrical tank of diameter 400 mm is bolted on the test bench of the test rig. The tank assembly includes a cross fixture mounted on its top to support an electromagnetic level sensor for measuring the MSWH as shown in Fig. 5. The MSWH data is acquired by the data acquisition system with a sampling rate of 100 Hz.

In order to capture the response of the tank, an accelerometer is pasted on the front face of the tank as shown in Fig. 5 with the axis of the accelerometer marked in direction of excitation of the tank. The output from the accelerometer (Kistler 8632C5) is recorded by using a VibPilot signal analyzer at a sampling rate of 2048 Hz.

The cylindrical tank is filled with water up to different heights of 0.2 m, 0.3 m, 0.4 m, and 0.5 m and sinusoidally excited in the lateral direction at frequencies from 1.2 Hz to 1.8 Hz in steps of 0.1 Hz with an amplitude of 1 mm for a time period corresponding to 30 cycles. The frequency span is decided in such a way that it includes the fundamental lateral natural frequency of sloshing liquid (Refer Table 3). Near the resonant frequency, additional trials in frequency step of 0.05 Hz are conducted for respective fill heights.

5. Results and discussion

The present section compares the frequency, MSWH, and acceleration of m_0 obtained from the present model against the experimentally observed values. Also, a comparison of force, exerted by the sloshing liquid on the tank wall, obtained from the present model is made with the sloshing force obtained from the Abramson et al. model [3].

Table 3
Natural frequency of the liquid column in the cylindrical tank for the first mode at different fill height.

Height(m)	Natural frequency (Hz)
h	f_0
0.2	1.47
0.3	1.51
0.4	1.51
0.5	1.51

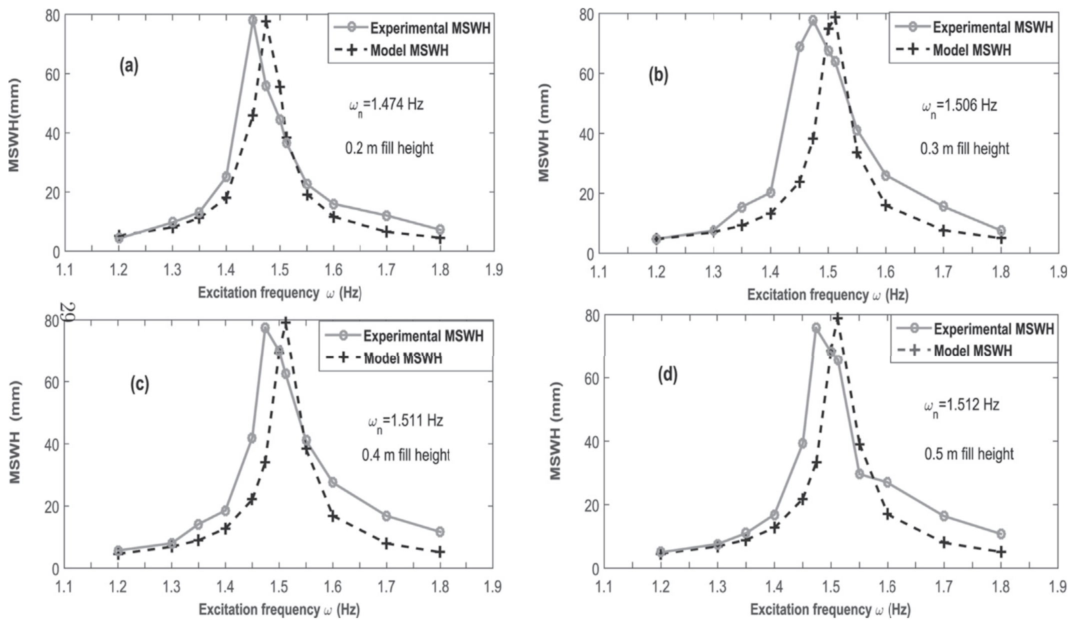

Fig. 6. Comparison of experimental and model MSWH for (a) 0.2 m (b) 0.3 m (c) 0.4 m (d) 0.5 m fill height.

5.1. Comparison of model and experimental MSWH

The natural frequency of the sloshing liquid is computed for each fill height, using Eq. (7) and is listed in Table 3. It is clear from Table 3 that natural frequency of sloshing liquid increases as the fill height to radius ratio increases and becomes constant as $h \geq 2R$. The MSWH, recorded using the level sensor, is compared with the response, x_1, of the sloshing mass, m_1, obtained from the present model.

Fig. 6 shows the comparison of MSWH obtained from the model and the experiment for all fill heights. It clearly indicates that the model resonance has occurred precisely at the natural frequencies computed in Table 3. In addition, the model and experimental MSWH are found to be in good agreement for pre-resonant frequencies as the free surface of liquid shows a planar behavior at the pre-resonant frequencies. As the excitation frequency approaches resonant frequency, the free surface loses its linear behavior. As a result, a difference is noticed in the experimental MSWH with respect to the model MSWH curve. On close examination of the MSWH obtained from the experimental result, it is noticed that the MSWH response is not symmetric with respect to the natural frequency. The response tends to show a left leaning nature as shown in Fig. 7. As result, the highest experimental MSWH occurs at a frequency lower than that predicted by the present model. This backward leaning behavior is similar to the response of a softening spring. This effect is more and more evident at lower fill heights (Fig. 7). This non-linear behavior of the sloshing mass results in small difference noticed in the model and experimental MSWH at post-resonance regions. The proposed model, being a linear model which does not include the rotational inertia of the sloshing mass as well as the spring softening effect, fails to capture these effects. A maximum difference of about 8 mm(approximately) is noticed between the highest MSWH observed experimentally and MSWH obtained from the model at the resonant frequency. Nonetheless, the pro-

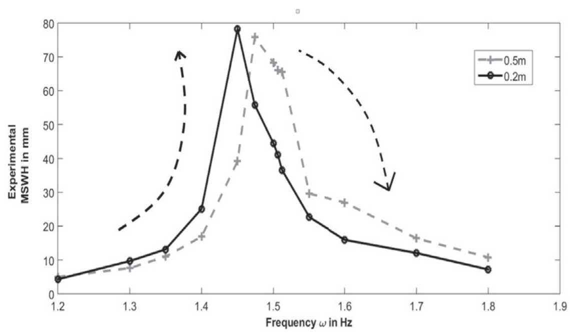

Fig. 7. The non-linear behavior of the free surface of liquid noticed in backward leaning experimental MSWH.

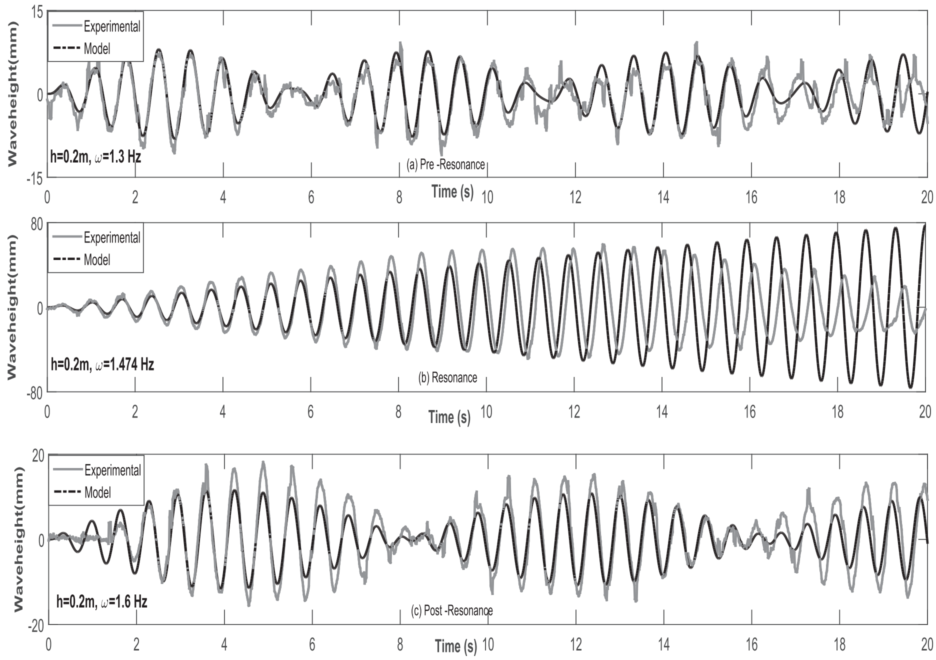

Fig. 8. Comparison of experimental MSWH and slosh mass displacement from the model at fill height of 0.2 m at (a) Pre-resonant frequency (b) Resonant Frequency (c) Post-resonant frequency.

posed model successfully captures amplitude of highest MSWH measured experimentally, at the resonant frequency, for all fill heights.

Figs. 8 and 9 gives the comparison of MSWH time histories obtained from model and experiment for the fill height of 0.2 m and 0.3 m respectively, for pre-resonance, resonance and post-resonance frequencies. Both the model and experimental MSWH

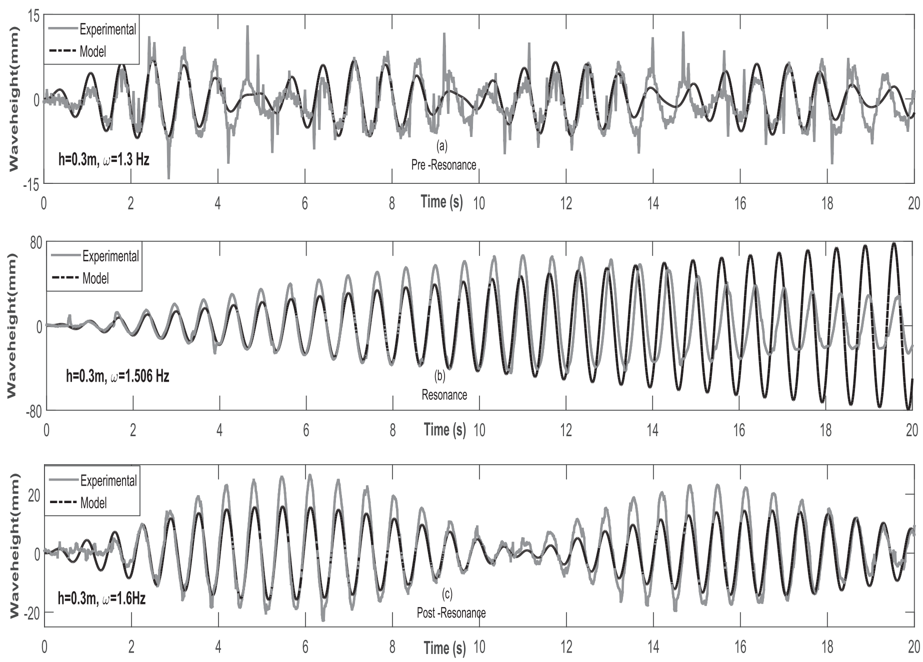

Fig. 9. Comparison of experimental MSWH and slosh mass displacement from the model at fill height of 0.3 m at (a) Pre-resonant frequency (b) Resonant Frequency (c) Post-resonant frequency.

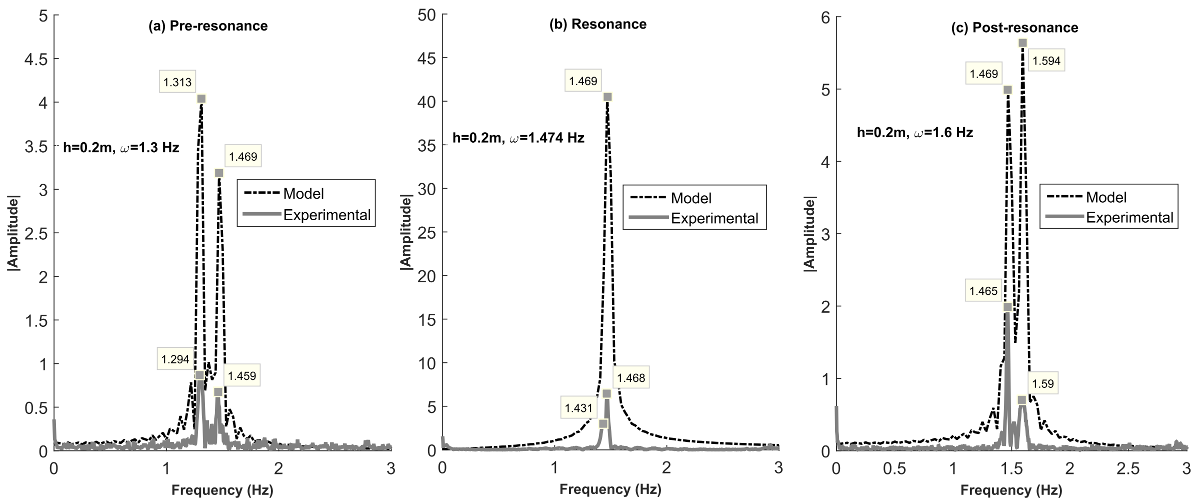

Fig. 10. Comparison of FFT responses of experimental MSWH and model slosh mass displacement at fill height of 0.2 m and excitation at (a) Pre-resonant frequency (b) Resonant Frequency (c) Post-resonant frequency.

signatures show beats for the pre-resonant and post resonant frequencies which is due to the fact that excitation frequency is close to the natural frequency of the sloshing liquid. MSWH obtained from the analytical model and that measured from experiment are in good agreement, in terms of amplitude and frequency, for the pre-resonant and post-resonance frequencies. A small difference is noticed in the amplitude of the experimental and model MSWH for the post resonant frequencies which is attributed to the local mode effects and formation of ripples associated with the free surface of the liquid at post-resonance frequencies.

Figs. 8(b) and 9(b) compares the MSWH signatures for fill height of 0.2 m and 0.3 m respectively, close to the resonance condition. Due to the inclusion of damping in the model, a finite amplitude of MSWH, is obtained from the present model which matches with the experimentally observed MSWH. This shows that the damping included in the analytical model is a true representation of the actual damping present in the system. The MSWH signature, obtained from the present model, shows an increasing trend as expected, at the resonant frequency of liquid. On the other hand, the experimental MSWH, increases and then reduces, showing a pattern of beats. In addition, a phase difference also can be noticed in both the model and experimental MSWH which is attributed to the shift in resonance between the model and experimental MSWH. Even though minor differences exist between the analytical model and experimental results, the primary objective of this study is to aid the designer in attaining quick results with respect to the MSWH and frequencies involved in lateral sloshing problem, when compared with the presently available CFD based methods, and hence to reduce the overall design cycle time.

In order to further investigate, a frequency spectrum analysis of both experimental and model MSWH data is carried out using FFT. Fig. 10 compares the spectrum for pre-resonance, resonance and post-resonance frequencies and at a fill height of 0.2 m. Fig. 10(a) and (c), show two clear peaks in the frequency spectrum of both model and experimental MSWH, which represent the excitation and resonant frequency of sloshing liquid and thus, explain the beats observed in the pre-resonance and post-resonance region. At resonance, the frequency spectrum of model MSWH data show only a single peak of 1.469 Hz, due to presence of a single resonant frequency. The experimental MSWH data, on the contrary, shows two separate peaks of 1.431 Hz and 1.468 Hz which illustrates the difference in amplitude and phase noticed in model and experimental MSWH time histories in Fig. 8b. This is because, in the experiment during sweeping, there is a hardware limitation on frequency increment, and one may not be able to excite the system exactly at the resonance. This disadvantage is not in there in mathematical model. This is also a reason for the difference in amplitude in the frequency domain. This Similar observation are noticed for the frequency spectrum analysis of model and experimental MSWH data for 0.3 m fill height in Fig. 11.

5.2. Comparison of tank accelerometer response

The response of the rigid mass, m_0, obtained from the present model is compared with the acceleration of tank measured during the experiment for the frequency range of interest, to further affirm the model accuracy. The response of tank is measured with the accelerometer pasted on the front face of the tank as shown in Fig. 5. Fig. 12 compares the peak to peak amplitude of acceleration measured experimentally and the response of rigid mass obtained from the proposed model for all fill heights

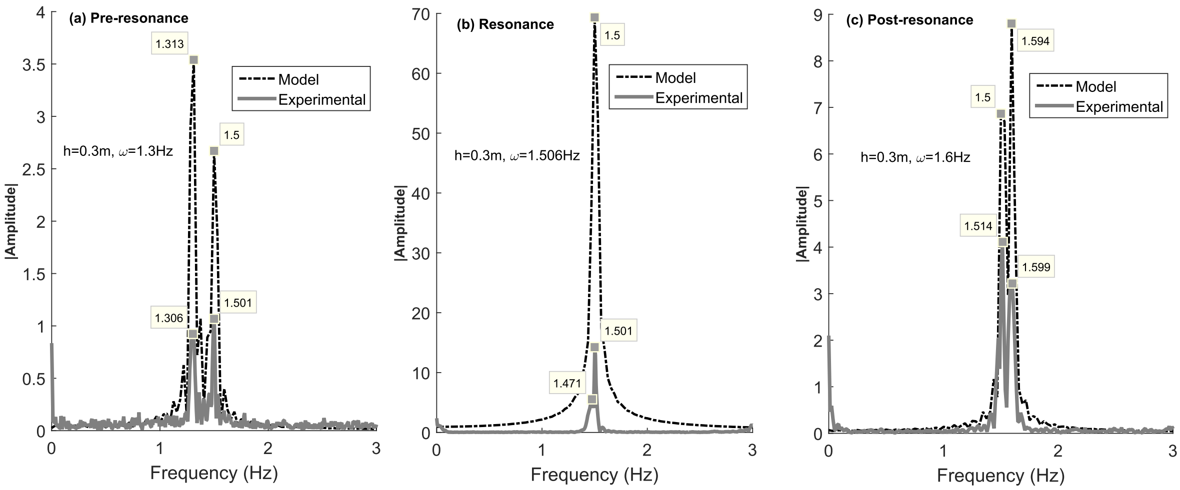

Fig. 11. Comparison of FFT responses of experimental MSWH and model slosh mass displacement at fill height of 0.3 m and excitation at (a) Pre-resonant frequency (b) Resonant Frequency (c) Post-resonant frequency.

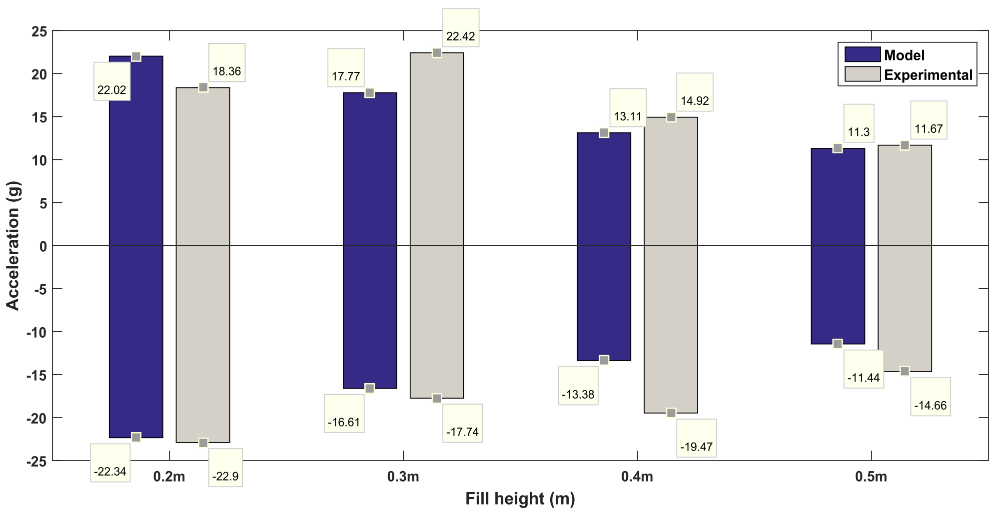

Fig. 12. Comparison of tank acceleration amplitude (peak to peak) observed during the experiment with the response of m_0 obtained from the model for all fill heights of liquid.

and at an excitation frequency of 1.4 Hz. The rigid mass response obtained from the proposed model is found to be in good agreement with the response of the tank obtained from the experiment as shown in Fig. 12, for all fill heights of sloshing liquid.

In addition to the above comparison, a frequency domain analysis is also carried to increase confidence in the present model. Fig. 13 compares the frequency peaks obtained from the FFT analysis of rigid mass acceleration data and tank accelerometer data for all fill heights. The spectrum of rigid mass acceleration data, for respective fill height contains two dominating peaks shown separately in Fig. 13(a) and (b) respectively. One peak corresponds to the excitation frequency of 1.375 Hz, recorded for all fill heights, and the other peak corresponds to the forced frequency response of tank which reduces from 537.1 Hz to 288.5 Hz, as the liquid fill level increases from 0.2 m to 0.5 m. These frequencies are present in the spectrum of the accelerometer response measured experimentally as well, for respective fill heights as shown in Fig. 13. These results confirm that response of the rigid mass is able to match the actual representation of the tank system including the non-sloshing portion of liquid and hence it can be stated that the present model is an accurate representation of the physical sloshing system.

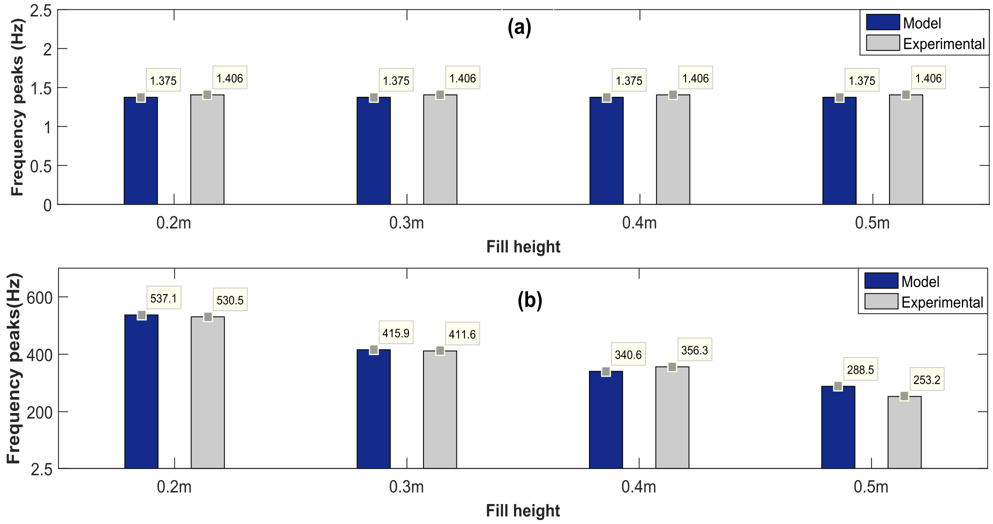

Fig. 13. Comparison of FFT analysis of acceleration (of mass m_0) obtained from the model and tank acceleration measured experimentally at an excitation frequency of 1.4 Hz (a) Peaks showing the excitation frequency (b) Peaks showing the forced frequency of tank.

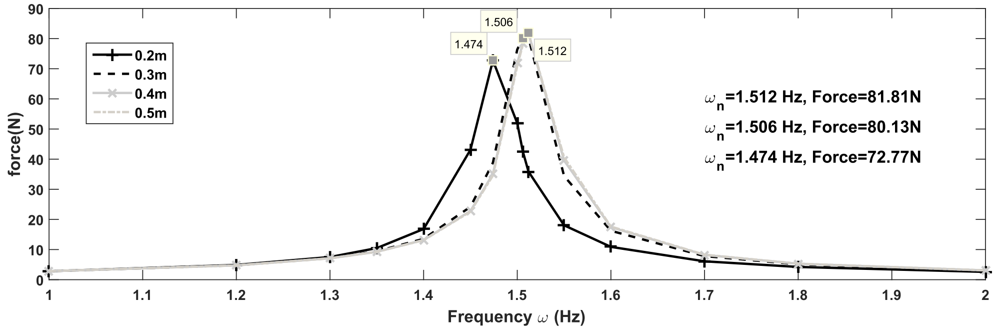

Fig. 14. Comparison of lateral sloshing force obtained from the present model for all fill heights.

5.3. Comparison of lateral sloshing force

The lateral force exerted by the sloshing liquid on the tank wall can be computed from the present model using Eq. (19). Fig. 14 shows the magnitude of lateral force obtained using the proposed model for all fill heights at excitation frequencies varying from 1.0 Hz to 2.0 Hz. It can be clearly observed from Fig. 14 that, for a particular fill height, the maximum lateral force occurs precisely at the respective resonant frequency. Also, as the fill height increases, the lateral force experienced by the container increases, due to fact that the stiffness and damping of sloshing mass increases with fill height. The lateral force curve for 0.4 m and 0.5 m lie over one another since the fundamental natural frequencies at these fill heights are almost equal.

In Fig. 15, the force experienced by the tank during sloshing which is calculated from model given by Abramson et al. using Eq. (11) is compared with same obtained from the proposed model for all fill heights. The present model, due to the inclusion of damping produces finite force amplitude at the resonant frequency.

Fig. 16 shows the comparison of lateral force time histories, obtained from the present model and Abramson model, at pre-resonance, resonance and post-resonance frequencies for a fill height of 0.5 m. It can be clearly observed from Fig. 16(a) and (c) that at pre-resonance and post-resonance frequencies, the lateral force obtained from the proposed model occurs as beats. This is expected since force exerted by the liquid indirectly depends on wave displacements of sloshing liquid, which also occurred as beats for these set of frequencies. On the other hand, the lateral force prediction by Abramson model reveals no beating pattern as shown in Fig. 16. Fig. 16(b) shows the force comparison for resonant frequency wherein the force obtained from the present model shows increasing trend which is expected, whereas the force obtained from Abramson model shows sinusoidal behavior with very high value due to the absence of damping.

Frequency spectrum of the force data obtained from the proposed model as well as Abramson model, for the fill height of 0.5 m, is shown in Fig. 17. The FFT of force obtained from the proposed model shows two distinct frequency peaks in Fig. 17(a)

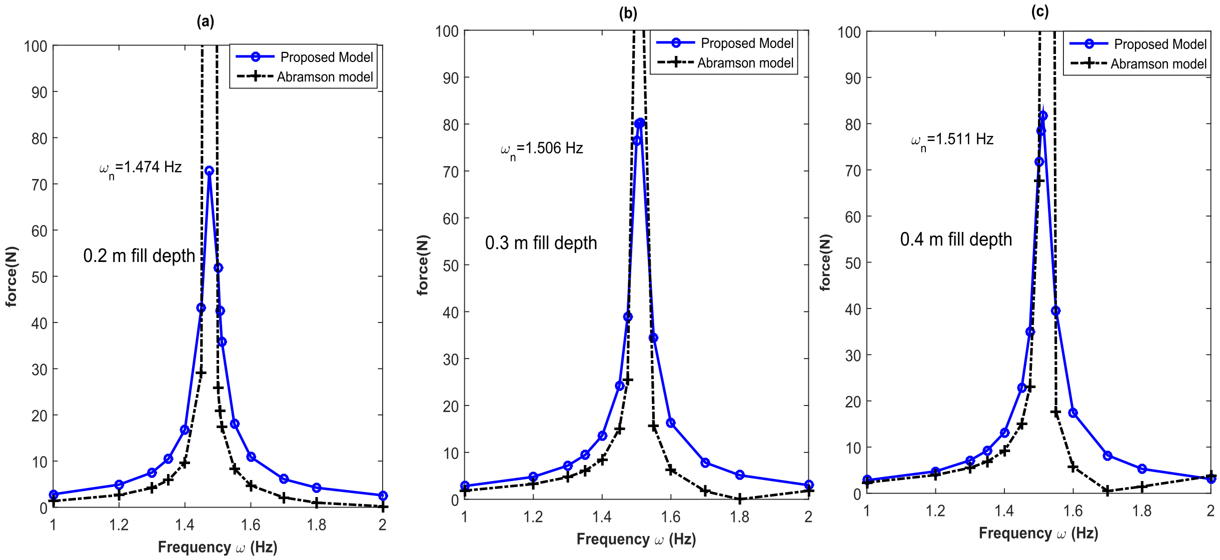

Fig. 15. Comparison of sloshing force, experienced by tank wall, obtained from the present model and model by Abramson et al. for (a) 0.2 m and (b) 0.3 m (c) 0.4 m fill height.

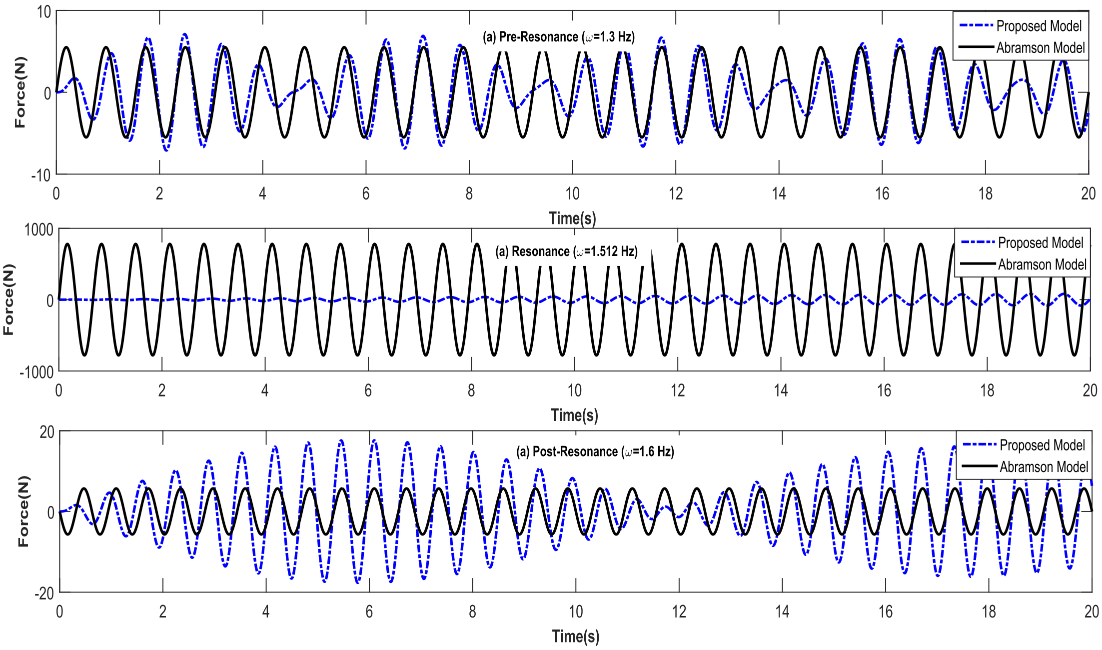

Fig. 16. Comparison of lateral force obtained from the present model and Abramson model for 0.5 m fill height at (a)Pre-resonance (b) Resonance (c) Post-resonance frequency.

and (c) for pre-resonance and post-resonance frequency respectively, which explains the beat phenomenon seen in force time histories in Fig. 16(a) and (c) respectively. These two distinct peaks represent the excitation and resonance frequency of liquid at the corresponding fill height of 0.5 m. On the other hand, the FFT of force data obtained from Abramson model shows a single peak of 1.313 Hz and 1.594 Hz in Fig. 17(a) and (c) respectively. Fig. 17(b) shows the frequency spectrum comparison at the resonant frequency which shows a single peak for both the models. It can be concluded from the force prediction results that the present model can predict sloshing forces more realistically when compared with those obtained from Abramson et al. [3] model.

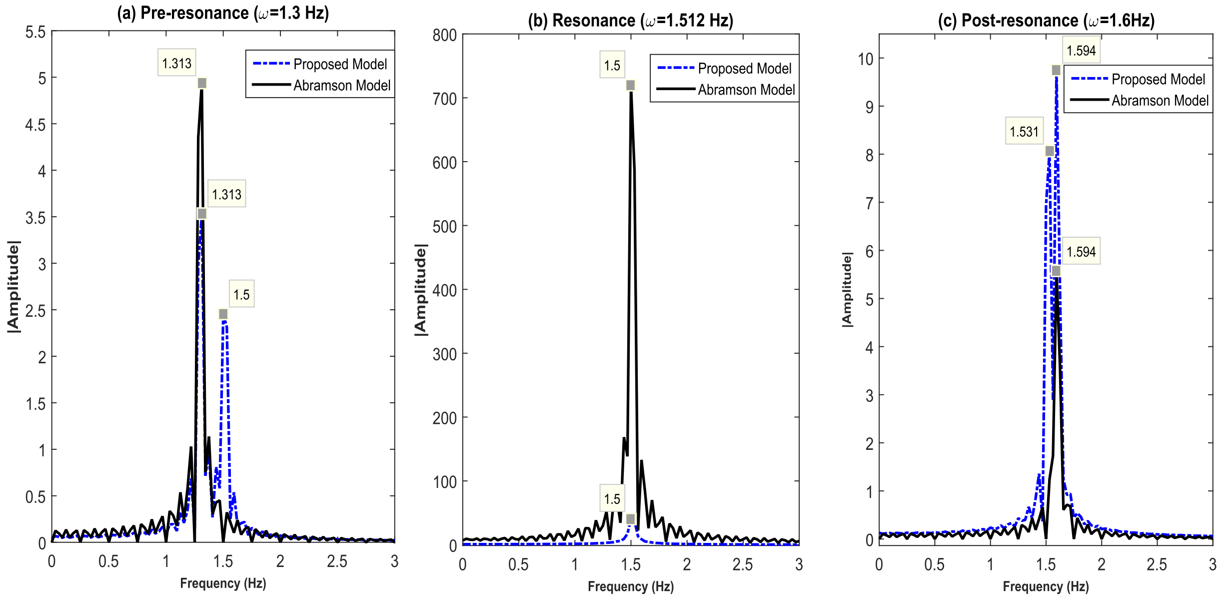

Fig. 17. Comparison of Frequency spectrum of Lateral force obtained from the present model and Abramson model for 0.5 m fill height at (a)Pre-resonance (b) Resonance (c) Post-resonance frequency.

6. Conclusions

Determination of MSWH is an inevitable part of design for engineering systems which carries liquid. The methods available at present, for eg. CFD based models and particle based methods like SPH, to estimate the MSWH, are cumbersome, expensive and requires expertise, which leads to increase in overall design cycle time. In this light, the present mechanical model, which is simple yet gives an innovative way to determine the MSWH and the force of sloshing liquid. This can be very effective in the preliminary design iterations of any lateral sloshing problem and will reduce the overall design cycle time. This model can prove to be a convenient tool in the hands of a design engineer while solving problems involving sloshing.

The parameters such as stiffness and damping of the rigid mass of the proposed model, are estimated with a simple hammer impact test. The proposed model imbibes the stiffness and mass parameters of the sloshing liquid from the mechanical model proposed by Abramson et al. [3] and transforms it to predict the waveform and amplitude of MSWH and sloshing force for the applicable frequency range of interest. It successfully simulates the damping present in the actual physical system and the beats observed in MSWH at pre-resonance and post-resonance frequencies.

The rigid mass response of the proposed model is found to be in good agreement with the experimentally measured accelerometer response. This again affirms the accuracy of the proposed model. The main highlight of the proposed model is its simplicity and ease of parameter estimation. The frequency content of both model and experimental MSWH matches reasonably well for pre-resonant and post-resonance frequencies. For, post-resonance excitations, a difference in amplitude of model and experimental MSWH is noticed which is due to the ripple formation and non-linearities observed in the free surface of the liquid. The prediction of lateral force and its comparison only with Abramson et al. model is provided. Experimental validation of the same is required for further confirmation of the prediction of lateral force. However, the present study is mainly aimed at aiding the design engineer in getting first hand results about the MSWH, force of sloshing liquid at the slosh frequencies and hence help in reducing the overall design cycle time.

The experimental study carried out proves that the proposed spring-mass-damper model is an accurate representation of water sloshing in a laterally excited cylindrical tank. There is ample scope for refinement and fine tuning of the present model to increase its accuracy in the post-resonance region in terms of MSWH amplitude prediction. This model can be enhanced by incorporating more modal masses, non-linear and torsional spring to capture the non-linear effects and rotation of the free surface near resonant frequencies. Also, the given model could be modified for predicting the MSWH in the case of different shaped containers and under different modes of excitations. Nonetheless, the methodology discussed paves the way for analytical prediction of sloshing wave height in a cylindrical tank during lateral excitation in an accurate and swift manner.

Acknowledgment

The authors extend their deep sense of gratitude and sincere thanks to Dr. Ashraf A.K., Shri Kodati Srinivas and Shri Sharath S. from Indian Space Research Organization (ISRO) for providing the slosh test facility and logistics support for conducting the experiments.

References

[1] K. Hatayama, Tsunami damage to oil storage tanks in the MW9.0 2011 tohoku, Japan earthquake, in: Proc. of ASME 2015 Pressure Vessels and Piping Conference, 2015.
[2] K. Yamamoto, M. Kawahara, Structural oscillation control using tuned liquid damper, Comput. Struct. 71 (4) (1999) 435–446.
[3] H.N. Abramson, H.W. Chu, G.E. Ransleben, Representation of fuel sloshing in cylindrical tanks by an equivalent mechanical model, ARS J. 31 (12) (1961) 1697–1705.
[4] M.A. Goudarzi, S.R. Yazdi, W. Marx, Seismic analysis of hydrodynamic sloshing force on storage tank roofs, Earthq. Spectra 26 (1) (2010) 131–152.
[5] S.S. Hough, The oscillations of a rotating ellipsoidal shell containing fluid, Proc. R. Soc. Lond. 186 (1) (1894) 299–301.
[6] K. Honda, M. Matsushita, Scientific Reports Tohoku, An Investigation of the Oscillations of Tank Water, vol. 21, Imperial University, 1913, pp. 131–148. First Series.
[7] H. Jeffries, Free oscillations of water in an elliptical lake, Proc. Lond. Math. Soc. 23 (1924) 455–476.
[8] B.M. Sen, Waves in canals and basins, Proc. Lond. Math. Soc. 26 (1927) 363–376.
[9] H.M. Westergaard, Water pressures on dams during earthquakes, Trans. Am. Soc. Civ. Eng. 98 (1933) 418–472.
[10] H. Lamb, Hydrodynamics, 6 edition, Dover publications, New York, 1945.
[11] J.J. Stoker, Water Waves, the Mathematical Theory with Applications (Wiley Classics Library), Interscience Publishers Inc., New York, 1957.
[12] J.W. Miles, On the Sloshing of Liquid in a Cylindrical Tank, Technical Report Report No. AM6-5, Ramo-Wooldridge Corp. Guided Missile Research Division, 1956.
[13] H.F. Bauer, W. Eidel, Free oscillations and response of a viscous liquid in a rigid circular cylindrical tank, Aero. Sci. Technol. 3 (8) (1999) 495–512.
[14] R.M. Cooper, Dynamics of liquids in moving containers, ARS J. 30 (8) (1960) 725–729.
[15] K. Kachigan, Forced Oscillations of a Fluid in a Cylindrical Tank, Technical Report ZU-7-046, Convair, 1955.
[16] P. Roberts, J. Chen, E. Basurto, Slosh Design Handbook, I - NASA-CR-406, NASA, Washington D.C., 1966.
[17] R. Buseck, H. Benaroya, Mechanical models for slosh of liquid fuel, in: Proceedings of the AIAA Aerospace Design Conference, AIAA, 1993.
[18] H.F. Bauer, Nonlinear mechanical model for the description of propellant sloshing, AIAA J. 4 (9) (1966) 1662–1668.
[19] R.W. Warner, J.T. Caldwell, Experimental Evaluation of Analytical Models for the Inertia and Natural Frequencies of Fuel Sloshing in Circular Cylindrical Tanks, Technical Report TN D-856, NASA, 1961.
[20] H.N. Abramson, The Dynamic Behavior of Liquids in Moving Containers, with Applications to Space Vehicle Technology, National Aerospace Space Administration(NASA), Special Publication SP-106, Washington D.C., 1966.
[21] S.K. Yalla, A. Kareem, Beat phenomenon in combined structure-liquid damper system, J. Eng. Struct. 23 (6) (2001) 622–630.
[22] N. Cavalagli, C. Biscarini, A.L. Facci, F. Ubertini, S. Ubertini, Experimental and numerical analysis of energy dissipation in a sloshing absorber, J. Fluids Struct. 68 (2017) 466–481.
[23] J. Anderson, S. Semercigil, O.F. Turan, A standing-wave-type sloshing absorber to control transient oscillations, J. Sound Vib. 232 (5) (2000) 839–856.
[24] M. Gradinscak, S.E. Semercigil, O.F. Turan, A sloshing absorber with a flexible container, in: T. Proulx (Ed.), Proceedings of the 28th IMAC, A Conference on Structural Dynamics, 2010, volume 3 of Conference Proceedings of the Society for Experimental Mechanics Series, Springer, 2010.
[25] M.A. Goudarzi, S.R. Yazdi, Investigation of nonlinear sloshing effects in seismically excited tanks, Soil Dyn. Earthq. Eng. 43 (2012) 355–365.
[26] G. Wu, Q. Ma, R.E. Taylor, Numerical simulation of sloshing waves in a 3d tank based on a finite element method, Appl. Ocean Res. 20 (6) (1998) 337–355.
[27] Rouf A. Ibrahim, Liquid Sloshing Dynamics: Theory and Applications, Cambridge university press, 2005.
[28] H.F. Bauer, Theory of Fluid Oscillations in Partially Filled Cylindrical Containers, Technical Report MTP-AERO-62-1, NASA-MSFC, 1962.
[29] W. Chu, Free surface condition for sloshing resulting from pitching and some corrections, ARS J. 30 (11) (1960) 1093–1094.
[30] H.F. Bauer, Tables and graphs of zeros of cross product bessel functions, Math. Comput. 18 (85) (1964) 128.
[31] G.N. Mikishev, N.Y. Dorozhkin, An experimental investigation of free oscillations of a liquid in containers, Izv Akad Nauk SSSR Otd Tekhn Nauk Mekh i Mashinostr(Russian) 4 (1961) 48–53.